10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 3.1 — Order and Equality of Matrices

Unit 3 | Solved Exercise | New Book (PECTAA)

Exercise 3.1 (Solved)

1. Write the number of rows and number of columns in each matrix.

A matrix is written in the form of rows (horizontal lines) and columns (vertical lines). Its order is always (number of rows) × (number of columns).

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Rows = 2, Columns = 2

Answer: Order = 2 × 2

$$B = \begin{bmatrix} 3 \\ 4 \\ 7 \end{bmatrix}$$

Rows = 3, Columns = 1

Answer: Order = 3 × 1

$$C = \begin{bmatrix} 8 & -10 & 11 \end{bmatrix}$$

Rows = 1, Columns = 3

Answer: Order = 1 × 3

$$D = \begin{bmatrix} 3 & 2 & 1 \\ 1 & 0 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Rows = 3, Columns = 3

Answer: Order = 3 × 3

$$E = \begin{bmatrix} 5 & 9 & -2 \\ -3 & 4 & 5 \end{bmatrix}$$

Rows = 2, Columns = 3

Answer: Order = 2 × 3

2. Write the order of each matrix.

$$A = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

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2 rows and 1 column

Answer: Order = 2 × 1

$B = \begin{bmatrix} 3 & 4 \end{bmatrix}$

1 row and 2 columns

Answer: Order = 1 × 2

$C = \begin{bmatrix} 3 & 5 \\ 5 & 6 \end{bmatrix}$

2 rows and 2 columns

Answer: Order = 2 × 2

$D = \begin{bmatrix} 0 & 3 & 4 \\ 5 & 0 & 9 \end{bmatrix}$

2 rows and 3 columns

Answer: Order = 2 × 3

$E = \begin{bmatrix} 0 & 1 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$

3 rows and 2 columns

Answer: Order = 3 × 2

$F = \begin{bmatrix} 5 \end{bmatrix}$

1 row and 1 column

Answer: Order = 1 × 1

3. Which of the following matrices are equal?

Two matrices are equal only when they have the same order and all their corresponding elements are equal. First simplify the entries, then compare.

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Given matrices

$$A = \begin{bmatrix} 0 & 3 \\ 4 & 1 \end{bmatrix} \quad (2 \times 2)$$

$$B = \begin{bmatrix} 9 \\ -7 \end{bmatrix} \quad (2 \times 1)$$

$$C = \begin{bmatrix} 6 & 1 \\ 4 & 2 \\ 8 & 3 \end{bmatrix} \quad (3 \times 2)$$

$$D = \begin{bmatrix} 9 \\ -7 \end{bmatrix} \quad (2 \times 1)$$

$$E = \begin{bmatrix} 6 & 1 \\ 4 & 2 \\ 8 & 3 \end{bmatrix} \quad (3 \times 2)$$

$$F = \begin{bmatrix} 0 & 3 \\ 4 & 1 \end{bmatrix} \quad (2 \times 2)$$

Simplified form

$$B = \begin{bmatrix} 9 \\ -7 \end{bmatrix} \text{ and}$$

$$D = \begin{bmatrix} 9 \\ -7 \end{bmatrix} \implies B = D$$

$$C = \begin{bmatrix} 6 & 1 \\ 4 & 2 \\ 8 & 3 \end{bmatrix} \text{ and}$$

$$E = \begin{bmatrix} 6 & 1 \\ 4 & 2 \\ 8 & 3 \end{bmatrix} \implies C = E$$

$$A = \begin{bmatrix} 0 & 3 \\ 4 & 1 \end{bmatrix} \text{ and}$$

$$F = \begin{bmatrix} 0 & 3 \\ 4 & 1 \end{bmatrix} \implies A = F$$

A and F have the same order (2 × 2) and the same entries

Answer: A = F, B = D, C = E

4. If a + 2, c − 3/b − 1, d + 4 = 5, 8/6, 4, then find a, b, c and d.

Equal matrices → compare the corresponding elements one by one.

a + 2 = 5

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a = 3

c − 3 = 8

c = 11

b − 1 = 6

b = 7

d + 4 = 4

d = 0

Answer: a = 3, b = 7, c = 11, d = 0

5. If 2x + 1, 4/7, 5 = 9, 4/7, y, then find x and y.

2x + 1 = 9

2x = 8

x = 4

y = 5

Answer: x = 4, y = 5

6. If a + b, 2d − 1/3b + 2, 4 = 10, 5/11, c, then find a, b, c and d.

3b + 2 = 11

3b = 9

b = 3

a + b = 10

a + 3 = 10

a = 7

2d − 1 = 5

2d = 6

d = 3

c = 4

Answer: a = 7, b = 3, c = 4, d = 3

7. If p + q, 5/11, p − 2q = 6, 5/11, 0, then find p and q.

p + q = 6 ... (1)

p − 2q = 0 ... (2)

From (2): p = 2q

Put in (1): 2q + q = 6

3q = 6

q = 2

p = 2(2)

= 4

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Answer: p = 4, q = 2

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